

Volcanoes

Deep inside Earth are pockets of hot, molten rock. Now and then, this fluid surges up to the surface and pours out in a volcanic eruption. Some volcanoes stay active over millions of years, whereas others erupt only for a few years.

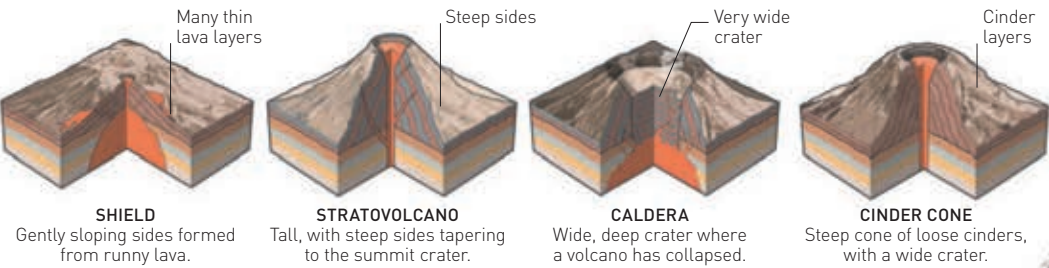
HOW A VOLCANO GROWS

A volcano builds itself up out of its own erupted materials, such as lava and ash. Red-hot when it flows out, this matter cools to form solid rock.



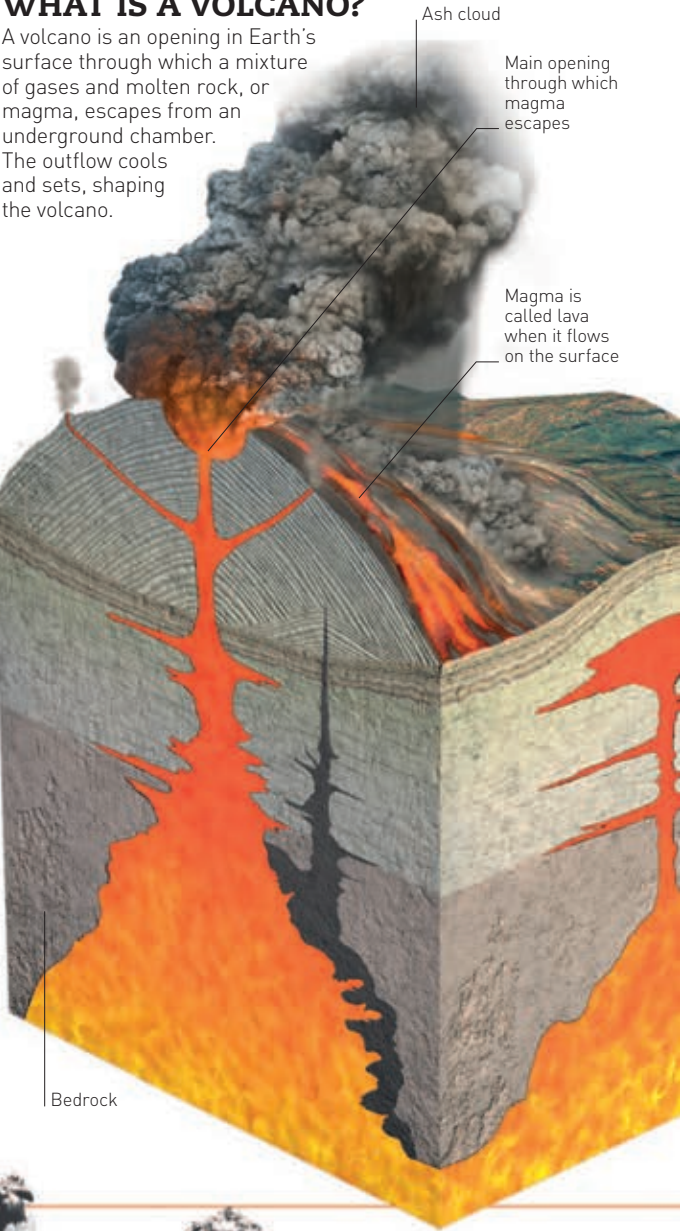
VOLCANO TYPES

Not all volcanoes have a steep “smoking mountain” shape. Other forms include shield volcanoes – which look like huge, upturned dinner plates – and small cinder cones. Calderas are craters that appear when a volcano collapses.



WHAT IS A VOLCANO?

A volcano is an opening in Earth’s surface through which a mixture of gases and molten rock, or magma, escapes from an underground chamber. The outflow cools and sets, shaping the volcano.

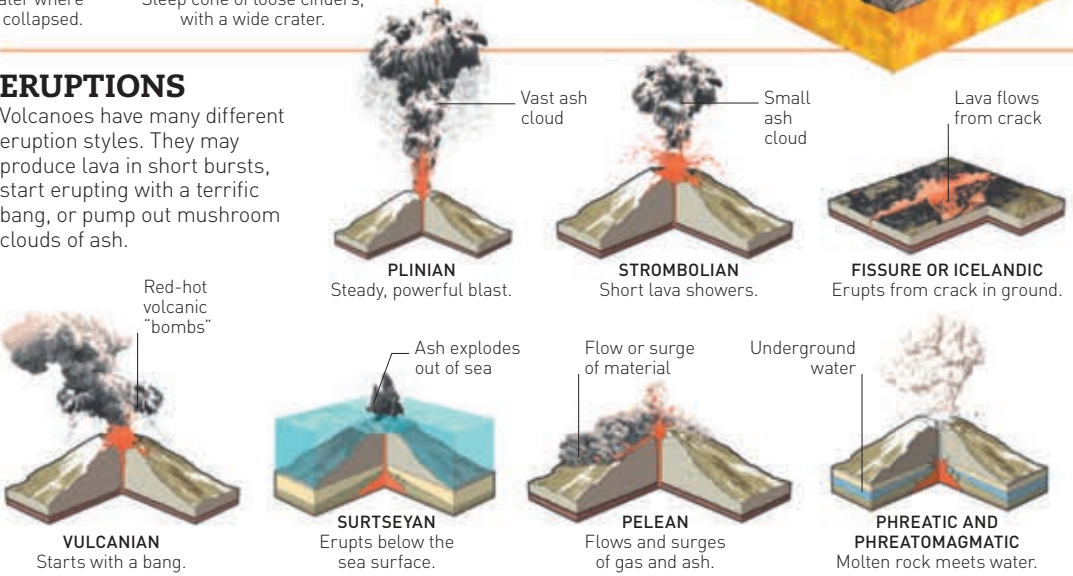


LARGEST ACTIVE SHIELD VOLCANOES

LOCATION	SHAPE	SUMMIT HEIGHT	MAXIMUM WIDTH OF BASE
Mauna Loa, Hawaii		4,169 m (13,677 ft)	95 km (59 miles)
Erta Ale, Ethiopia		613 m (2,011 ft)	80 km (50 miles)
Sierra Negra, Galápagos		1,500 m (4,921 ft)	50 km (31 miles)
Nyamuragira, Democratic Republic of Congo		3,058 m (10,033 ft)	45 km (28 miles)
Kilauea, Hawaii		1,247 m (4,091 ft)	50 km (31 miles)

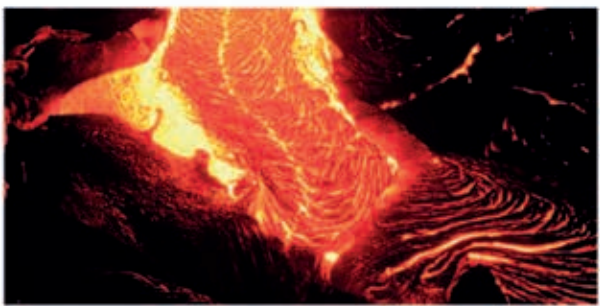
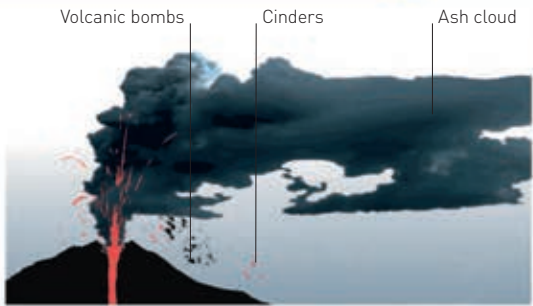
ERUPTIONS

Volcanoes have many different eruption styles. They may produce lava in short bursts, start erupting with a terrific bang, or pump out mushroom clouds of ash.



VOLCANIC FALLOUT

An eruption blasts a lot of dangerous material into the air. Molten lava “bombs”, hot cinders, rocks, and ash fly upwards then fall to the ground. Poisonous, suffocating gases are also given off.



LAVA

The red-hot flow that pours from a volcano is lava – the name given to molten rock, or magma, once it reaches the surface. The hottest lavas are thin and runny, and flow a long way before cooling and solidifying. Others are thick and sticky, and creep just a short way before coming to a stop.